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Review Article

The Beneficial Effects of L. reuteri Probiotics in Development of the Toddlers' Immune Systems and Gut Health in Bangladesh

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Abstract: A well-researched probiotic bacteria called Lactobacillus reuteri (L. reuteri) has the ability to colonize several animals. L. reuteri can be found in the skin, breast milk, urinary tract, gastrointestinal tract, and other parts of the human body. Different people have varying amounts of L. reuteri. There are a number of advantages to L. reuteri. First, L. reuteri has the ability to create compounds that are antimicrobial, such as organic acids, ethanol, and reuteri. L. reuteri's antimicrobial activity enables it to prevent the colonization of pathogenic bacteria and alter the composition of the host's commensal microbiota. Second, L. reuteri can strengthen the immune system of the host. For instance, some L. reuteri strains can enhance the formation and activity of regulatory T cells while suppressing the production of pro-inflammatory cytokines. Third, because L. reuteri has the possibility to fortify the intestinal barrier, its colonization may lessen the number of bacteria that move from the gut lumen to tissues. It has been proposed that microbial translocation across the gut epithelium causes inflammation to start. Therefore, boosting the colonization of L. reuteri may help to alleviate inflammatory illnesses, including those that affect the gut as well as distant organs. Notably, during the past few decades, L. reuteri abundance in humans has declined, and at the same time, inflammatory illness occurrences in toddlers have increased. For the correct developmental growth of toddlers, obvious supplementation, or prebiotic modification of L. reuteri may be an alluring preventative and/or therapeutic pathway counter to inflammatory illnesses.

Keywords: Lactobacillus Reuteri, Microbiota, Probiotic, Immune System, Inflammatory Diseases, Toddler, Gut Health